



Basic Information

- Experiment No: 14
- Date: 17/09/2026
- Title: Implementation of Runtime Polymorphism using Method Overriding



Objective / Aim

To write a Java program that demonstrates runtime polymorphism using inheritance, method overriding, and a base class reference to calculate the rate of interest of different banks.



System Requirements

- Hardware: Standard PC setup (Core i3/i5 Processor, 4GB/8GB RAM).
- Software: Windows/Linux/macOS, Java Development Kit (JDK 17 or JDK 21), IntelliJ IDEA / Eclipse / NetBeans / VS Code.

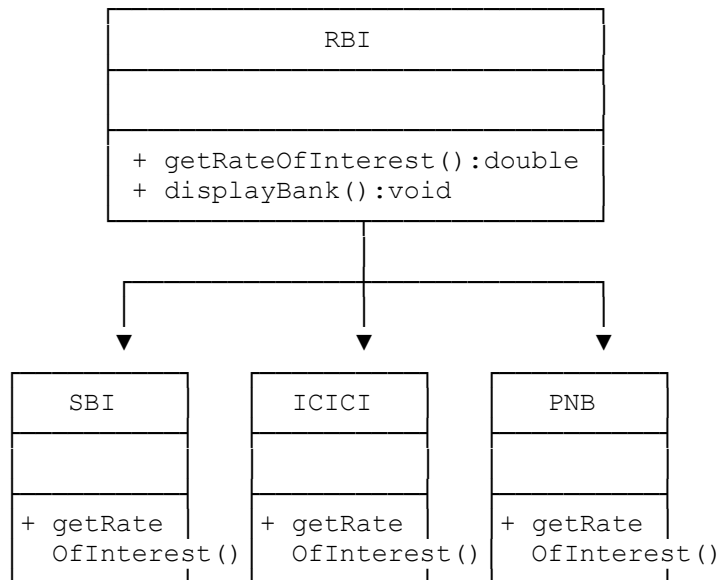
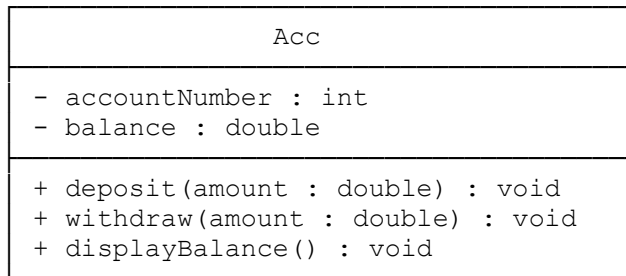
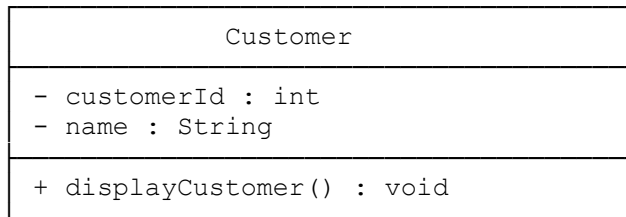


Algorithm / Logic Description

1. Start the program.
2. Create a `Customer` class with customer ID and name.
3. Create an `Acc` class with account number and balance.
4. Define methods for depositing, withdrawing, and displaying balance.
5. Create a base class `RBI` with a method `getRateOfInterest()` returning 4.0%.
6. Create derived classes `SBI`, `ICICI`, and `PNB` that inherit from `RBI`.
7. Override the `getRateOfInterest()` method in each derived class.
8. Create a `Scanner` object to read the bank name from the user.
9. Declare an `RBI` reference variable.
10. Create the appropriate bank object based on the entered bank name.
11. Call `displayBank()` using the `RBI` reference.

12. Demonstrate runtime polymorphism through method overriding.
13. Display the bank's rate of interest.
14. Stop the program.

UML Class Diagram





Source Code

```
package record;

import java.util.Scanner;
// Customer class
class Customer {
    int customerId;
    String name;
    Customer(int customerId, String name) {
        this.customerId = customerId;
        this.name = name;
    }
    void displayCustomer() { // displayCustomer method to display the customer details
        System.out.println("Customer ID : " + customerId);
        System.out.println("Customer Name : " + name);
    }
}
// Account class to display the customer bank details
class Acc {
    int accountNumber;
    double balance;
    Acc(int accountNumber, double balance) {
        this.accountNumber = accountNumber;
        this.balance = balance;
    }
    void deposit(double amount) {
        if (amount > 0) {
            balance += amount;
            System.out.println("Deposited : " + amount);
        }
    }
    void withdraw(double amount) {
        if (amount > 0 && amount <= balance) {
            balance -= amount;
            System.out.println("Withdrawn : " + amount);
        } else {
            System.out.println("Insufficient balance");
        }
    }
    void displayBalance() {
        System.out.println("Balance : " + balance);
    }
}
// RBI is the base class
```

```

class RBI {
    double getRateOfInterest() {
        return 4.0;
    }
    void displayBank() {
        System.out.println("RBI rate of interest is : "+ getRateOfInterest() + "%");
    }
}
// SBI derived class SBI inherited RBI
class SBI extends RBI {
    @Override
    double getRateOfInterest() {
        return 7.0;
    }
}
// ICICI derived class ICICI inherited RBI
class ICICI extends RBI {
    @Override
    double getRateOfInterest() {
        return 6.5;
    }
}
// PNB derived class PNB inherited RBI
class PNB extends RBI {
    @Override
    double getRateOfInterest() {
        return 6.0;
    }
}
// Main class
public class BankROI {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print( "Enter the Bank name to find the rate of Interest : ");
        String bankName = sc.nextLine();
        // RBI reference variable
        RBI rbi;
        if (bankName.equalsIgnoreCase("RBI")) {
            rbi = new RBI();
        }
        else if (bankName.equalsIgnoreCase("SBI")) {
            rbi = new SBI();
        }
        else if (bankName.equalsIgnoreCase("ICICI")) {
            rbi = new ICICI();
        }
        else if (bankName.equalsIgnoreCase("PNB")) {

```

```

        rbi = new PNB();
    }
    else {
        System.out.println("Invalid bank name");
        sc.close();
        return;
    }
    // Runtime polymorphism
    rbi.displayBank();
    sc.close();
}
}

```

Compilation & Execution Commands

Save the file as:

BankROI.java

Compile the code:

```
javac -d . BankROI.java
```

Run the compiled bytecode:

```
java Excersice.BankROI
```



Output Screen

```

"C:\Program Files\Java\jdk-24\bin\java.exe" "-javaagent:C
Enter the Bank name to find the rate of Interest : SBI
RBI rate of interest is : 7.0%

Process finished with exit code 0

```